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Title,Authors,Year,DOI or URL,Abstract,Source
"Object-centred automated compliance checking: a novel, bottom-up
approach","Omar Doukari; David Greenwood; Kay Rogage; Mohamad
Kassem","2022","https://doi.org/10.36680/j.itcon.2022.017","Building
Information Modelling (BIM) is changing how built assets are delivered and
operated. A built asset is represented as a set of objects, each with an
identity, attributes, and relations. This object-oriented nature enables new
approaches for ensuring compliance with a range of requirements (e.g.,
industry guidelines; project and client-specific requirements; and building
codes and standards). Based on an adapted version of a simulated annealing
concept, this paper proposes an automated compliance checking classification
and identifies a set of desired characteristics these methods should fulfill.
It then demonstrates a bottom-up object-centred approach for automated model
checking and the corresponding plugin prototype. To demonstrate the
feasibility and accuracy of the approach, two case studies were processed
using existing BIM object libraries, and results show significant time
savings compared to traditional manual methods [1] [2] ","Journal of Information
Technology in Construction (ITcon) [3] "
"Rule capture of automated compliance checking of building requirements: a
review","Zijing Zhang; Ling Ma; Tim Broyd","2023","https://doi.org/10.1680/
jsmic.23.00005","In the architecture, engineering and construction (AEC)
industry, building design needs to be checked against regulations before it
can be finalized and progress to construction. The traditional manual
compliance checking process is error-prone and time-consuming. As a solution,
automated compliance checking (ACC) was proposed. Rule capture is a crucial
bottleneck of ACC. Despite many studies in this domain, no research has
synthesized the themes or identified future research opportunities. This
paper conducts a systematic literature review focusing on rule capture in ACC
and identifies challenges in the field. Findings reveal that the current rule
representation development process lacks a methodological framework, cannot
represent “unknowns” or “side-effects,” struggles with ambiguous rules, and
is often limited by rule engines or data models [4] [5] ","Proceedings of the
ICE – Smart Infrastructure and Construction [6] "
"NLP-based Automated Compliance Checking of Data Processing Agreements
against GDPR","Orlando Amaral Cejas; Muhammad Ilyas Azeem; Sallam Abualhaija;
Lionel Briand","2023","https://doi.org/10.1109/TSE.2023.3288901","When a data
processor differs from the data controller, their processing agreement (DPA)
must comply with the EU GDPR. Manual checking of DPA compliance is
challenging due to the convoluted language and ambiguity of legal texts. This
paper proposes an automated solution to check a given DPA against GDPR
requirements. We first extract “shall” requirements from GDPR relevant to
DPAs and compile a glossary of legal concepts. Then, leveraging natural
language processing (NLP), we automatically generate phrasal representations
of the DPA’s text and compare them to predefined representations of the GDPR
requirements. The approach assesses GDPR compliance and recommends missing
information. In evaluations on 30 real DPAs, the approach detected 618 of 750
true violations with 76 false positives, achieving ~89.1% precision, 82.4%
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recall, and 84.6% accuracy (improving to ~94% with minimal manual verification) [7] [8] ","IEEE Transactions on Software Engineering [9] [10] "
"Automated code compliance checking research based on BIM and knowledge graph","Junlong Peng; Xiangjun Liu","2023","https://doi.org/10.1038/s41598-023-34342-1","Automated code compliance checking plays an important role in advancing the construction industry. While traditional drawing reviews rely on human experts, this study aims to achieve automatic code review using Building Information Modeling (BIM) and knowledge graph technology. We use knowledge graphs to transform textual code provisions into a structured, computer-recognizable format via natural language processing. BIM model data are exported and converted to a Resource Description Framework, after which compliance rules are applied to generate a review report. An example demonstrates the method’s feasibility, effectively addressing the manual dependency and inefficiency of current review processes [11] [12] ","Scientific Reports (Nature) [13] "
"Semi-automatic representation of design code based on knowledge graph for automated compliance checking","Mingsong Yang; Qin Zhao; Lei Zhu; Haining Meng; Kehai Chen; Zongjian Li; Xinhong Hei","2023","https://doi.org/10.1016/j.compind.2023.103945","Automated compliance checking (ACC) verifies whether designs comply with building codes. Interpreting complex design code provisions into a computer-processable form is a major challenge. This study presents a semi-automated approach to represent design codes using a knowledge graph, facilitating ACC. It combines manual and automated techniques for rule capture, aiming to reduce reliance on intensive expert involvement. A case study demonstrates that the proposed knowledge graph-based representation can effectively encode structural design code rules, enabling an ACC system to check structural calculations in openBIM format [14] [15] ","Computers in Industry [16] "
"Semantic web-based automated compliance checking with integration of Finite Element analysis","Panagiotis Patlakas; Ioannis P. Christovasilis; Lorenzo Riparbelli; Franco K.T. Cheung; Edlira Vakaj","2024","https://doi.org/10.1016/j.aei.2024.102448","Automatic compliance checking (ACC) is a promising approach to ensure designs meet building and planning regulations, especially with Building Information Modeling (BIM). However, little work has integrated computational engineering methods like Finite Element Analysis (FEA) into ACC. This paper proposes a holistic approach incorporating FEA into the Industry Foundation Classes (IFC) schema and using a Semantic Web-based system for ACC. We introduce the Building Compliance Ontology (BCO) to capture all aspects of regulatory compliance, focusing on modeling regulatory statements rather than just building components. In a case study on a modular steel structure’s design code compliance, we enrich an IFC model with FEA results and implement ACC checks using SHACL. The approach automatically verifies compliance by executing SHACL rules in a Semantic Web environment [17] [18] ","Advanced Engineering Informatics [19] "
"Automated Compliance Checking System for Structural Design Codes in a BIM Environment","Wonhui Goh; Jaeguk Jang; Sang I. Park; Bong-Hyuck Choi; Heuiseok Lim; Goangseup Zi","2024","https://doi.org/10.1007/s12205-024-1121-5","This study proposes a framework for an automated compliance checking (ACC) system to verify structural calculations in an openBIM model. The framework has four main components: (1) classification of design codes by structural component and limit state; (2) preparation of a

structural model using Industry Foundation Classes; (3) translation of code provisions into a machine-readable rule language; and (4) design of an ACC system focusing on rule flow for structural design codes. We demonstrate the framework with a case of checking the strength and serviceability of a prestressed concrete bridge girder. Using ifcOWL, SWRL, and a Drools inference engine, the ACC system successfully verified the bridge design. The proposed framework can guide development of ACC systems in structural engineering and can be customized for various design codes and data sources [14] [15] ","KSCE Journal of Civil Engineering [20] [21] "

"Automated Building Information Modeling Compliance Check through a Large Language Model Combined with Deep Learning and Ontology","Nanjiang Chen; Xuhui Lin; Hai Jiang; Yi An","2024","https://doi.org/10.3390/buildings14071983","Ensuring compliance with complex industry standards and regulations during building design is challenging with manual methods. This paper leverages natural language processing and deep learning to improve rule interpretation and compliance checking in the BIM domain. However, traditional approaches require extensive feature engineering and large labeled datasets. We propose an innovative automated compliance checking framework integrating a large language model (LLM), deep learning classifiers, and an ontology-based knowledge model. The LLM’s few-shot learning capability reduces the need for large annotated datasets, and a deep learning model pre-classifies regulatory text to improve the LLM’s information extraction accuracy. This combination automates processing of regulatory texts with minimal manual intervention, providing a scalable and adaptable solution for compliance checking in construction [22] [23] ","Buildings (MDPI) [24] [25] "

"Moving automated compliance checking to the operational phase of the building life-cycle: analysis and feasibility study in the UK","Thomas Beach; Jonathan Yeung; Ali Ghorghi; Yacine Rezgui","2024","https://doi.org/10.1080/15623599.2024.2366727","Construction regulations ensure baseline safety for built assets, but formal compliance checks currently occur only at certain stages (design, construction, etc.). This paper explores extending compliance checking throughout the operational phase of a building’s life-cycle by leveraging digitization and automation. We propose using automated and semi-automated data capture and analysis to perform continuous compliance checks in operation. The feasibility is evaluated through literature review, regulatory analysis, and proof-of-concept tests. Results indicate that whole-life compliance checking is increasingly important to monitor high-risk assets, but purely manual inspections are impractical and costly for operational checks. There are many viable use cases and available technologies to implement this approach, highlighting the potential for improved safety and maintenance through automated operational compliance monitoring [26] [27] ","International Journal of Construction Management [28] "

"Automating Compliance Evidence Extraction with Machine Learning","Sunday O. Olatunji; Abdulrahman S. Almuhaidib; Abdulaziz S. Alhuwaishil; Ahmad S. Alarfaj; Khalifah W. Alzwaimel; Hassan A. Alsaleem; Abdulaziz A. Alsudais; Naeem M. Owaida; Fahd A. Alhaidari; Sghaier R. Chabani","2025","https://doi.org/10.1007/978-981-97-8588-9_3","This paper reviews the intersection of cybersecurity and machine learning in automating compliance evidence extraction. It highlights the potential of automation to improve efficiency and reliability in compliance processes. The study focuses on how machine

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learning algorithms can analyze documents to determine if they meet
prescribed evidence standards, addressing the challenge of manually reviewing
voluminous compliance documentation. It surveys nine recent studies,
examining the machine learning methods, datasets, and outcomes for automating
compliance checks. The paper concludes that using machine learning for
compliance assessment can save time, reduce costs, and improve accuracy in
cybersecurity compliance evaluations [29] [30] ","Proceedings of the 1st Int.
Conf. on Creativity, Technology, and Sustainability (CCTS 2024) [31] "
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https://www.researchgate.net/publication/383130607_Automated_Compliance_Checking_System_for_Structural_Design_Codes_in_a_BIM_Environment

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